increase loan to value ratios. When liquidity tightens, though, too many lenders rely on asset sales, causing asset prices and loan recovery to plummet. Because lenders do not account for the effects of their lending on the 'fire sale' price, and subsequently on lending by others, they may have an excessive incentive to build leverage. These effects are exacerbated if, over time, lenders become reliant on asset sales for recovery, rather than on upfront project evaluation and due diligence. Another possible channel is that banks themselves become more levered, or equivalently, acquire more illiquid balance sheets, if the central bank signals it will intervene in a sustained way when times are tough because unemployment is high.

Leverage need not be the sole reason why exit may be volatile after prolonged unconventional policy. Investment managers may fear underperforming relative to others. This means they will hold a risky asset only if it promises a risk premium (over safe assets) that makes them confident they will not underperform holding it. A lower path of expected returns on the safe asset makes it easier for the risky asset to meet the required risk premium, and indeed draws more investment managers to buy it – the more credible the forward guidance on 'low for long', the more the risk taking. However, as investment managers crowd into the risky asset, the risky asset is more finely priced so that the likelihood of possible fire sales increases if the interest rate environment turns. Every manager dumps the risky asset at that point in order to avoid being the last one holding it.

Leverage and investor crowding may therefore exacerbate the consequences of exit. When monetary policy is ultra-accommodative, prudential regulation, either of the macro or micro kind, is probably not a sufficient defence. In part, this is because, as Fed Governor Stein so succinctly put it, monetary policy 'gets into every crack', including the unregulated part of the financial system. In part, ultra